Charge Below, Gravity Above:
RealQM, Maxwell, and Emergent Gravitation

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Written in respectful cooperation between Claes Johnson and Claude (Anthropic), with Claude contributing a balanced, neutral view.

Abstract

RealQM together with Maxwell's equations describes matter and light from charge and energy alone: the electron is a charge cloud in ordinary three-dimensional space, inertia is the derived quantity E/c^2, and mass enters nowhere as a primitive [1]. Gravitation is absent from this description — and need not be added. Gravity is negligible microscopically (between two protons it is $\sim 10^{-36}$ of the electric force) and dominates only at large scales, where electromagnetism has screened itself away in neutral bulk matter while unscreened gravity accumulates. We argue that this clean split — electromagnetic, charged and small *below*; gravitational, neutral and large *above* — is naturally read as gravity being *emergent*: a coarse-grained, collective phenomenon rather than a fundamental force acting between individual particles, in the spirit of Sakharov's induced gravity [5], Jacobson's derivation of the Einstein equation from thermodynamics [6], and the holographic emergence of spacetime from entanglement [8, 7]. The resulting picture is economical: *one* mechanics (Newton's second law $\mathbf{F} = \dot{\mathbf{p}}$), *one* inertia ($E/c^2$), and *two* forces separated by regime — electromagnetism coupling to charge, gravity to energy. On this view inertial mass (E/c^2, the energy content) is fundamental while *gravitational* mass — the role of that energy as gravity's charge — is emergent, which turns the observed equality of the two masses from an unexplained coincidence into a consequence of emergence. We are explicit about the limits. Emergent gravity emerges as *general relativity*, not Newtonian gravity (Newton is only its weak-field shadow); the proposal is established only in holographic models, not proven for our universe; and the framework is confined to non-extreme physics.

1 Two forces, two regimes

Ordinary physics is run by two long-range forces, and they occupy almost disjoint territories. Electromagnetism governs the small and the charged: atoms, molecules, chemistry, materials, light. Gravitation governs the large and the neutral: apples, planets, stars, galaxies. The boundary between them is not arbitrary; it follows from two facts, one on each side.

First, *gravity is extraordinarily weak*. The gravitational attraction between two protons is about 10^{-36} of their electrostatic repulsion; between two electrons, about 10^{-43}. At the scale of an atom gravity is not merely small, it is irrelevant to any digit anyone measures. Second, *electromagnetism screens, and gravity does not*. Charge comes in two signs and bulk matter is neutral, so over macroscopic bodies the electric forces cancel; mass comes in one sign, so gravity never cancels — it *accumulates*. Aggregate enough neutral matter and electromagnetism has quietly removed itself while gravity has been adding up, until, at planetary scale, the weakest force in nature is the only one left standing.

So the division "electromagnetic below, gravitational above" is not a modelling convenience but the way the two forces are actually organised: weakness keeps gravity out of the microscopic world, and screening keeps electromagnetism out of the macroscopic one.

2 One inertia, one mechanics, two forces

Underneath the two regimes there is a single mechanics. RealQM, together with Maxwell's equations, describes the microscopic world with *no mass as a primitive* [1]: the energy of matter is computed from charge and a geometric scale, and when a charge moves its inertia is carried by the momentum of its own field. Maxwell's equations supply the mechanism: the field momentum $\mathbf{g} = \mathbf{E} \times \mathbf{B}/4\pi c$ of a moving charge integrates to $\mathbf{p} \propto (\text{energy}/c^2)\,\mathbf{v}$ — J. J. Thomson's electromagnetic mass of 1881 [4] — while the exact law,

$$\mathbf{p} = \frac{E}{c^2}\,\mathbf{v}, \tag{1}$$

with total energy and unit coefficient, is the measured relation of the Penning trap [1] (the bare field integral alone gives the classical factor 4/3). Either way the inertial "mass" is the energy content E/c^2. Newton's second law, in its original form $\mathbf{F} = \dot{\mathbf{p}}$, then governs the response, with the force supplied by electromagnetism, $\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$, which couples to *charge* and contains no mass at all.

The macroscopic regime uses the *same* law and the *same* inertia. A neutral body has energy E, hence inertia E/c^2, and Newton's second law moves it; the only addition is a second force, gravity, which couples not to charge but to that same energy E/c^2. The economy is worth stating plainly:

	couples to	inertia (response)
electromagnetism	charge q	E/c^2
gravitation	energy E/c^2	E/c^2

One mechanics ($\mathbf{F} = \dot{\mathbf{p}}$), *one* inertia ($E/c^2$), *two* forces reading two different properties of the same matter — its charge and its energy. Nothing here needs a separate primitive called mass; "mass" is the name of the inertia, and the inertia is energy.

The two properties differ in one decisive respect, and it governs everything that follows: *charge comes in two signs, energy in one.* Because charge is $\pm$ it can cancel — unlike charges attract, like charges repel, and bulk matter is neutral — so electromagnetism *screens*. Because the gravitational charge is energy, and the total energy of real matter is positive (potential energy may be negative, but $E_{\text{total}} = mc^2 > 0$), gravity's source has a single sign: it cannot cancel, it can only *accumulate*, and it is always *attractive*. This single fact is the root of the whole separation below — screening removes electromagnetism over neutral bulk while single-signed gravity piles up — and of the character of each force: $\pm$ sources make electromagnetism a spin-1 field with attraction and repulsion, a single-signed attractive source makes gravitation even-spin (spin-2, once light-bending excludes spin-0). A one-signed charge that only ever adds is, moreover, exactly the kind of quantity whose effect is collective and large-scale, which is why gravity, alone among the forces, looks emergent.

3 The nuclear scale: charge and energy, contracted

The fundamental layer reaches further down than the atom. RealNucleus [3] applies the *same* functional — charge clouds minimising Coulomb energy against a localisation cost — to the *nucleus*, with only the geometric scale contracted from the ångström and electron-volt of chemistry to the fermi and MeV of nuclear binding. One theory, one scale smaller, reproduces nuclear binding as Coulomb geometry: the He-4/deuteron binding ratio comes out ≈ 13 against an experimental ≈ 12.7, and the burning of hydrogen to helium appears as a rearrangement of protons and electrons rather than the action of a separate force. On this reading the "strong" nuclear binding is not a new fundamental interaction but RealQM at the nuclear scale — charge and energy once more, contracted by the coefficient of the localisation term. Electromagnetism and nuclear binding become *one theory at two scales*, joined by that single geometric coefficient; the fundamental layer of the picture then spans the nuclear and the atomic worlds alike, and only *gravity* is left to emerge above them. (This is a strong and unconventional claim; the limits section below records what it has still to answer.)

Two scales, one relation. A cloud's size varies inversely with its geometric scale while its energy varies directly with it, so size and mass are linked: a contracted cloud costs more localisation and Coulomb energy and is therefore *heavier*, an extended one *lighter*. Electron and proton sit at the two ends of this relation — the electron big and light (atomic scale, ~ 0.5 Å, 0.511 MeV), the proton small and heavy (nuclear scale, ~ 1 fm, 938 MeV) — so "proton small, electron big" and "proton heavy, electron light" are one fact, not two. What RealQM does *not* supply is the *value* of the ratio: the proton/electron size ratio $\sim 5 \times 10^4$ is the nuclear-scale datum, the same input as the mass ratio m_p/m_e, which no theory derives from first principles. RealQM explains the size difference as a scale difference and ties it to the mass difference; fixing the scale itself is the mass-spectrum problem, part of the frontier below, and it is left open.

Its *existence*, however, is no idle fact. Were the atomic and nuclear scales equal, there would be no compact nucleus distinct from a diffuse electron cloud — no layered atom, no chemistry, no molecules, no structure. The separation of scales is a *precondition for complexity*, so any universe that supports structure, and hence observers, must exhibit it. This is an anthropic rationale for *why* the two scales are separated, not a derivation of *how far*: the magnitude of the ratio remains open, but its necessity for a complex world does not.

4 Why the separation is clean

Because the two forces read different properties and inhabit different scales, they do not interfere in the regimes where each operates. In the microscopic world gravity is 10^{-36} too weak to enter RealQM's Coulomb bookkeeping, so RealQM+Maxwell is *complete without gravitation*: atoms, bonds, spectra, and the propagation of light are electromagnetic through and through. In the macroscopic world the constituent charges have screened to neutrality, so gravitation is *complete without electromagnetism*: celestial mechanics needs no Maxwell field. The separation is therefore not a truncation we impose but a consequence of *weakness* on one side and *neutrality* on the other.

There is exactly one regime where the split must fail: where both conditions break at once, i.e. where gravity is strong *and* energy is unscreened. That is the province of astrophysics and cosmology — light bending near a star, the field energy of radiation gravitating in the early universe — and there electromagnetism and gravitation genuinely couple through their shared currency, energy. Everywhere else, across the whole of non-extreme physics, the two theories run in disjoint domains.

5 Gravity as emergent on larger scales

That gravity should be at once the weakest force, unscreenable, universal in its coupling to energy, and important only in the large invites a stronger reading than "fundamental but faint": that gravity is *emergent* — a collective, coarse-grained effect of many degrees of freedom, in the way that temperature and pressure are collective effects meaningless for a single molecule. This is not a fringe suggestion but a recurring theme of modern gravitational theory.

Sakharov proposed already in 1967 that gravity is an "induced" elastic response of the quantum vacuum [5]. Jacobson showed in 1995 that Einstein's field equations can be *derived* as a thermodynamic equation of state, from the Clausius relation $\delta Q = T\,dS$ applied to local causal horizons [6] — gravity appearing not as a fundamental force but as the large-scale thermodynamics of whatever the underlying degrees of freedom are, much as the gas law follows from statistics without tracking molecules. Padmanabhan developed this thermodynamic route in detail [7], and in the holographic setting the emergence is sharpest of all: bulk spacetime geometry, and hence gravity, is built from the quantum *entanglement* of a lower-dimensional field theory [8].

On this reading the microscopic world has no gravity to speak of — not merely a negligible amount, but no fundamental gravitational interaction between individual charges at all, just as a single molecule has no temperature. Gravity is what *emerges* when enough energy is coarse-grained together. This fits the two-regime picture exactly: the fundamental layer is charge and energy (RealQM+Maxwell), and gravitation is the large-scale collective phenomenon that appears above it. The very features that made the separation clean in §3 — gravity's weakness, its non-screening, its universal coupling to energy — are precisely what one expects of an emergent, statistical effect rather than a microscopic force.

6 Gravitational mass: an emergent role of a fundamental quantity

A corollary sharpens what is, and is not, emergent, and it concerns mass directly. The *quantity* E/c^2 — the energy content of a piece of matter — is fundamental and present at every scale: it is the inertial mass, RealQM computes it for a single atom, and a lone electron carries it. What is emergent is the *role* of that quantity as *gravitational* mass, the charge that sources and feels gravity. Where there is no gravity — the microscopic world — E/c^2 is inertia and nothing more; gravitational mass is not merely negligible there, it has *no referent*, exactly as a single molecule has no temperature. Gravitational mass appears only when gravity does, at large scales. So inertial mass is fundamental; gravitational mass is the emergent role it takes on once gravity has emerged.

This has a welcome consequence: it *explains* the equivalence principle rather than positing it. The equality of gravitational and inertial mass — the deepest empirical fact about gravity, tested to 10^{-13} — is ordinarily an unexplained coincidence. Here it is automatic. Emergent gravity arises as the collective response to the energy that is *already present* as inertia, so it can couple to nothing but E/c^2: the gravitational charge *is* the inertial energy, by construction and not by accident. Equivalence is then not an input but an output of emergence.

7 What is honest to claim, and what is not

The picture is attractive and, in its non-extreme domain, coherent; but three limits must be stated plainly rather than glossed.

First, emergent gravity emerges as general relativity, not Newton. Jacobson's construction yields Einstein's equations, and holography yields general relativity in the bulk. A consistent large-scale gravity that couples to energy and propagates at finite speed is, in any case, forced towards general relativity by the requirement that the gravitational field's own energy also gravitate. Newtonian gravity is only the weak-field, slow-motion corner of what emerges; it is the working form in §2, but not the fundamental large-scale theory, which must reproduce light bending, perihelion precession and gravitational waves.

Second, emergence is proven only in models, not for our world. The thermodynamic and holographic derivations are compelling and, in anti-de Sitter/conformal-field-theory examples, essentially established; but that gravity is emergent *in our universe* remains a hypothesis. The standard framework still treats gravity as a fundamental (spin-2) field, and the two views are not yet experimentally separated.

Third, the framework is bounded to non-extreme physics, and inherits open sectors. With RealNucleus the *strong*/nuclear sector is brought inside — as charge clouds at a contracted scale rather than a separate force — but this is a heterodox reduction that has still to be reconciled with the quark–gluon evidence of quantum chromodynamics (deep-inelastic scattering, jets, asymptotic freedom); the *weak* interaction (beta decay) remains more open in the programme; the elementary rest energies remain inputs, as in every theory; and the genuinely relativistic and quantum-field regimes — pair creation, the precision anomalies of quantum electrodynamics, and quantum gravity itself — lie outside by construction. Naming that boundary is a strength: it excludes exactly the regimes where the clean separation is known to break.

8 A theory of (non-extreme) everything?

It is worth asking plainly whether the assembled picture is a theory of everything. The pieces do fit into one frame. A single fundamental layer — *charge and energy*, governed by RealQM — runs from the nucleus (RealNucleus) through the atom, with Maxwell for the field; one interaction, Coulomb geometry, does the work at both the nuclear and atomic scales, so that electromagnetism and nuclear binding are the same theory at two settings of the geometric scale. Above this layer gravitation *emerges* as the single-signed, cumulative, large-scale role of the same energy. One mechanics ($\mathbf{F} = \dot{\mathbf{p}}$), one inertia ($E/c^2$), one organizing asymmetry (charge $\pm$ screens, energy $+$ accumulates): from these the whole of non-extreme physics — chemistry, materials, optics, nuclear binding, mechanics, and gravity — is meant to follow. As a *unified framework for non-extreme physics*, this is a genuine and economical candidate.

Whether it deserves the unqualified name "theory of everything" is a separate matter, and honesty requires the qualifier. Four things remain to be answered before the word is earned. The strong sector is brought in by RealNucleus, but as a heterodox reduction that must still confront the quark–gluon evidence of quantum chromodynamics. The weak interaction — beta decay, flavour — is not yet accounted for. The elementary rest energies (the ratios m_e/m_p and the rest) are inputs, as in every theory, not outputs. And the deepest grail, a *quantum* theory of gravity unifying it with the rest, lies outside: the framework's gravity is classical and emergent, established in holographic models but not proven for our world. To these must be added the extreme regimes — high-energy pair creation, the precision anomalies of quantum electrodynamics — excluded by construction.

The honest verdict is therefore this. A *complete* theory of everything is neither claimed nor in hand. But a *unified framework for the non-extreme world* — charge and energy from the nucleus to the atom, gravity emergent above, and a single inertia E/c^2 carried through all of it — is coherent, economical, and, within its declared bounds, self-consistent. That is a real unification. The frontier beyond it — the weak force, the mass spectrum, quantum gravity, and reconciliation with the strong-interaction evidence — is precisely where the word "everything" would have to be won.

9 Conclusion

For the vast, non-extreme middle of physics — chemistry, materials, optics, mechanics, the motion of ordinary bodies — a single economical picture suffices. The fundamental layer is *charge and energy* — RealQM from the nucleus (RealNucleus) to the atom, with Maxwell for the field — inertia the derived quantity E/c^2 and no primitive mass. Upon it act two forces reading two properties of the same matter: electromagnetism its charge, gravitation its energy, kept in disjoint domains by the weakness of gravity below and the neutrality of bulk matter above. And the most natural place for gravitation in this picture is not as a second fundamental force but as an *emergent*, large-scale, collective effect — appearing, like thermodynamics, only when enough energy is gathered together, and absent as a microscopic interaction between charges. What emerges must, to match the sky, be general relativity rather than Newton, and whether it truly emerges in our universe is still open; but the organising thought is clean and, within its declared bounds, self-consistent: *charge and energy below, emergent gravity above, and a single inertia E/c^2 carried through both.*

References

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